Essential Computer Science (I)

Moving from classical computing to quantum computing

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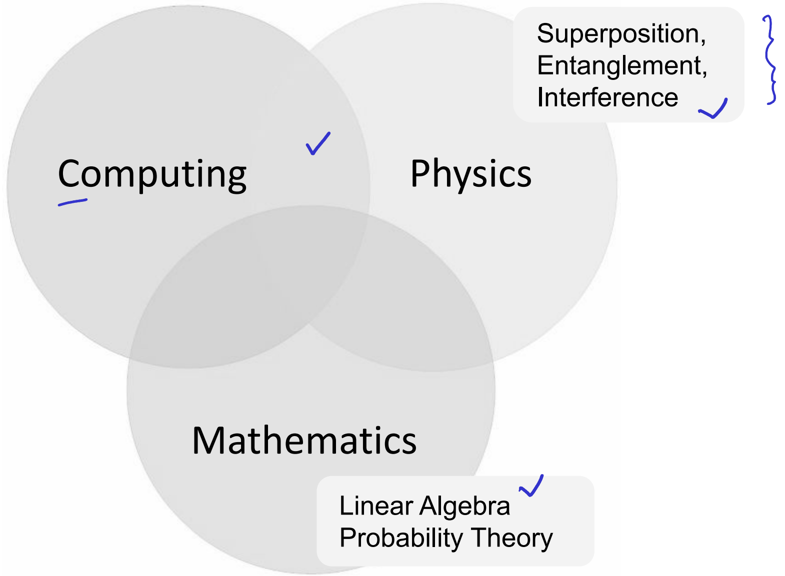
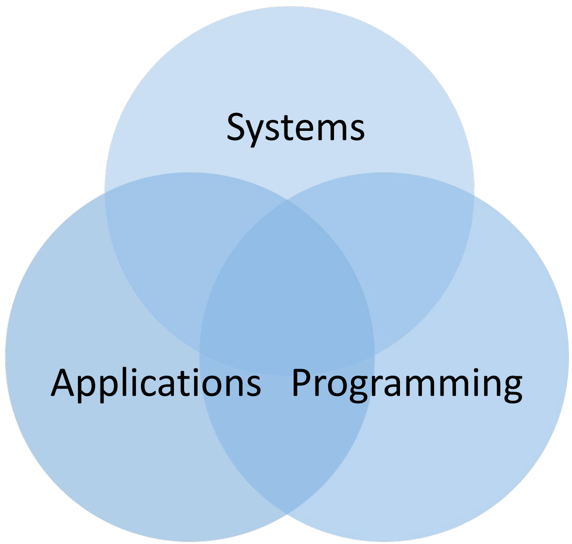
What are we covering in this module?

Theoretical Computer Science (I)



Practical Implementations



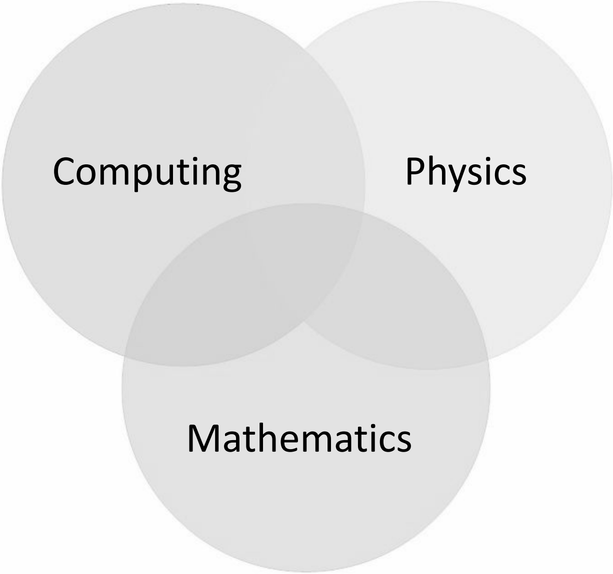
This module will be conceptual and not strictly mathematical. The aim is to cover a broad overview of **necessary** tools needed later in the course.

- Physical aspects of computation
- Mathematical model for computation
- Theoretical framework for computation
- Computational problems, algorithms, analysis

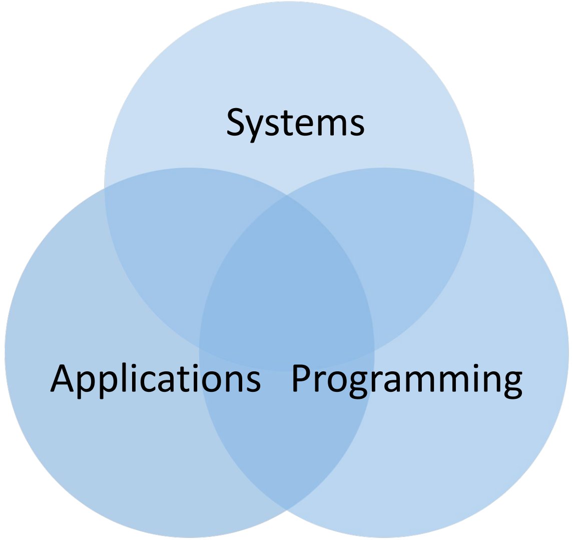
- System-level and architecture-level
- Quantum programming
- Quantum applications layer
- End-to-end quantum toolchain

What are we covering in this module?

Theoretical Computer Science (I)



Practical Implementations



- Compare and contrast between classical computing and quantum computing
  - classical bits/algorithms and quantum bits/algorithms
  - classical complexity theory and quantum complexity theory
  - classical software and quantum computational software
  - classical devices and quantum computational devices
- Understand the bridge between theoretical frameworks and practical implementations